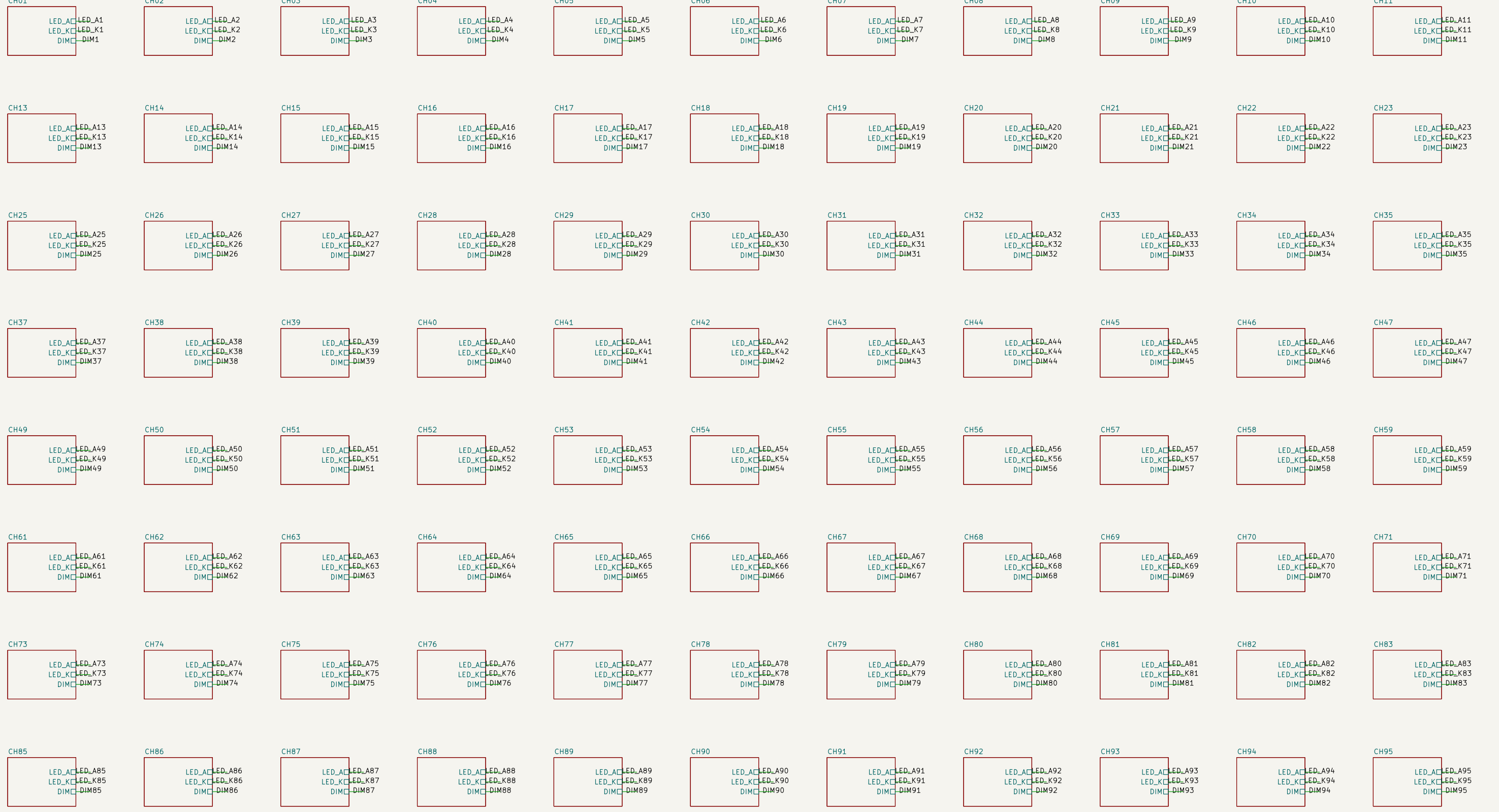
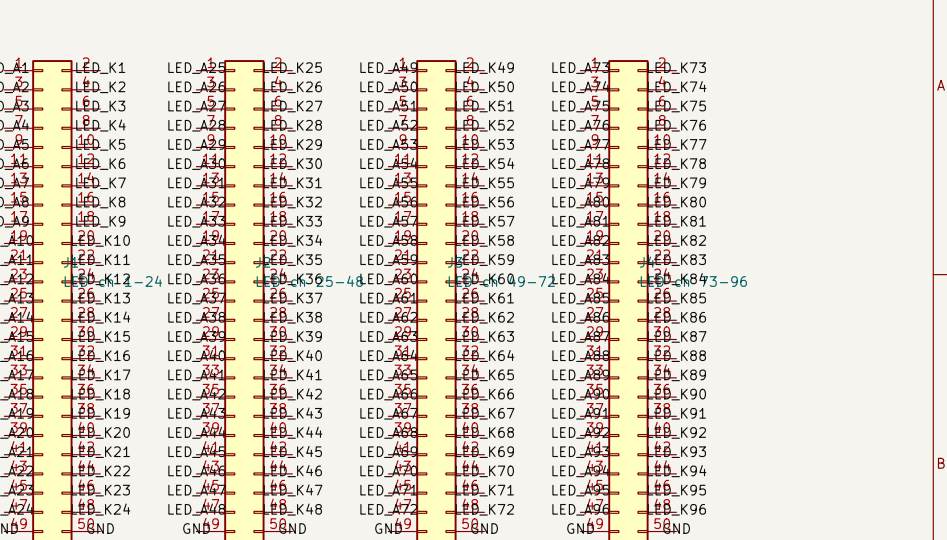


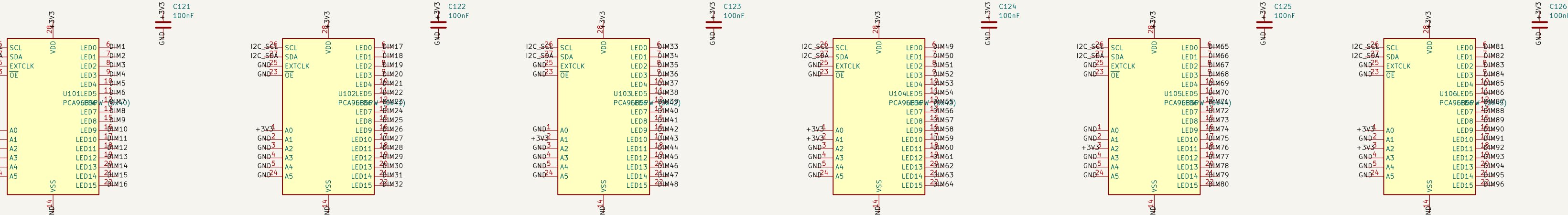
96 DRIVER CHANNELS (CH01..CH96) – see channel.kicad_sch



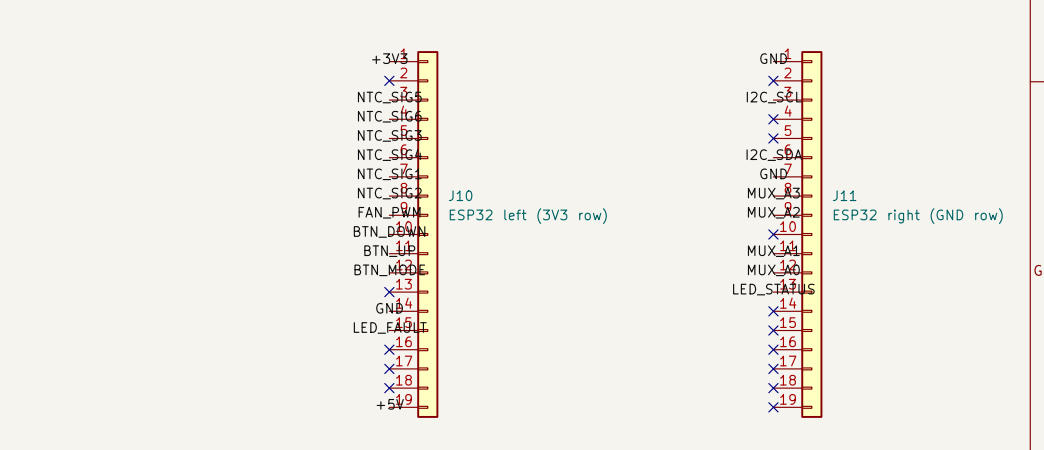
RIBBONS TO LED BOARD



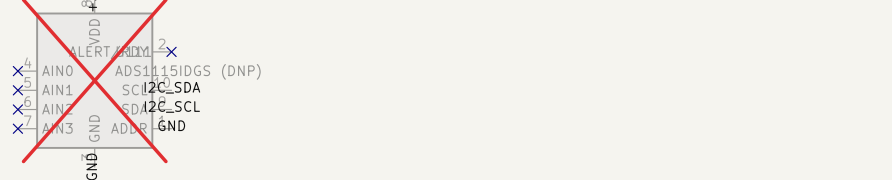
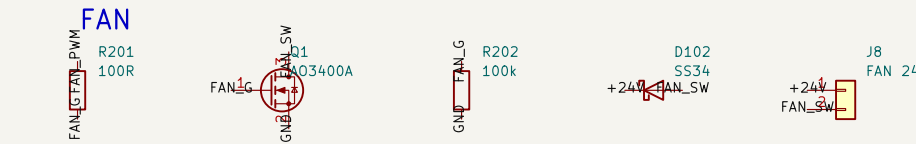
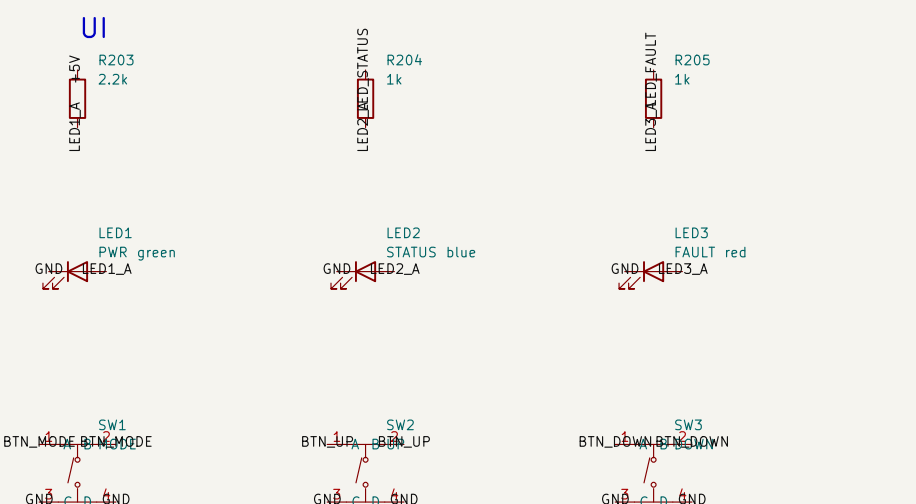
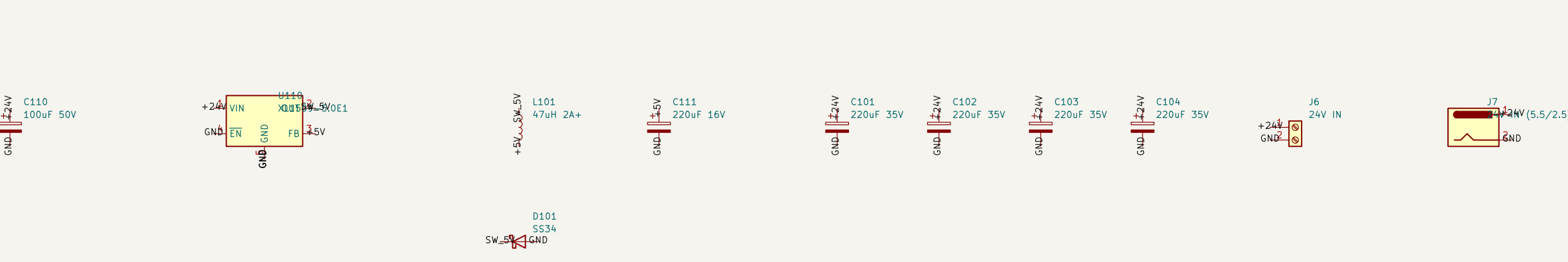
PCA9685 PWM BANK 0x40–0x45



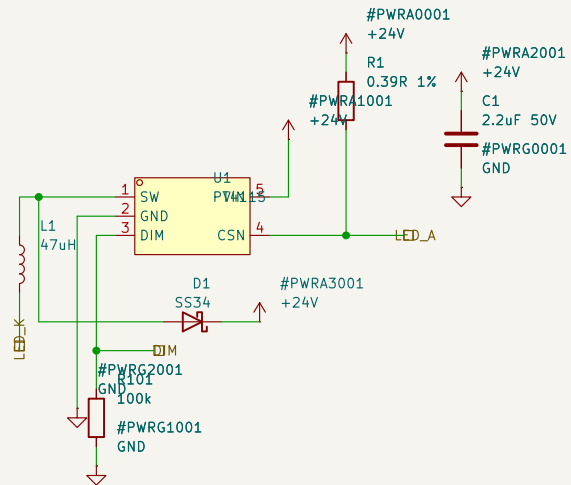
ESP32 DEVKITC–38 SOCKET (J10 left / J11 right, pin1 = 3V3/GND end)



5V RAIL + 24V INPUT



One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH01/

File: channel.kicad_sch

Title: Channel: PT4115 buck 256mA

Size: A4

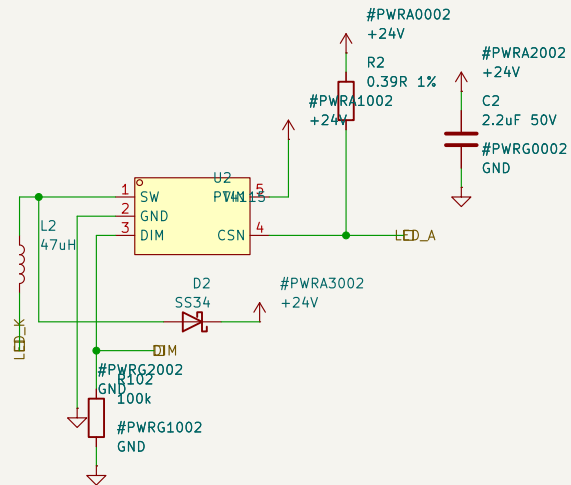
Date: 2026-07-13

Rev:

KiCad E.D.A. 10.0.3

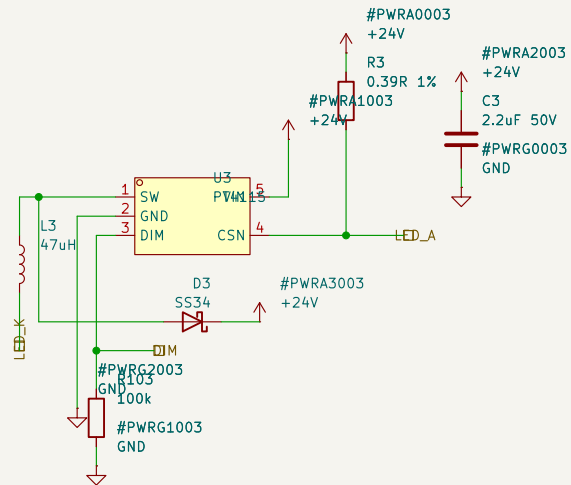
Id: 2/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



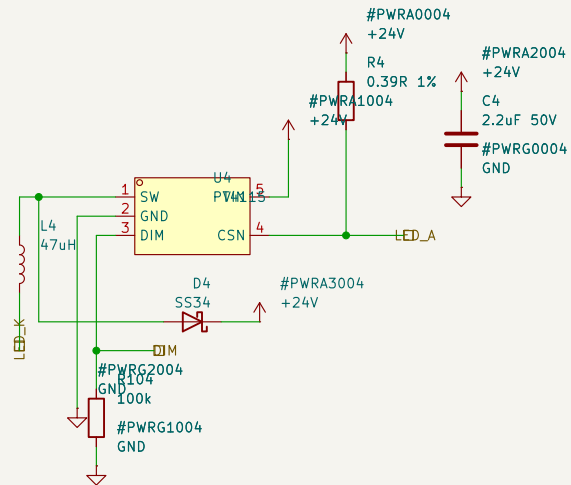
Sheet: /CH02/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3	Id: 3/97	

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH03/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 4/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH04/

File: channel.kicad_sch

Title: Channel: PT4115 buck 256mA

Size: A4

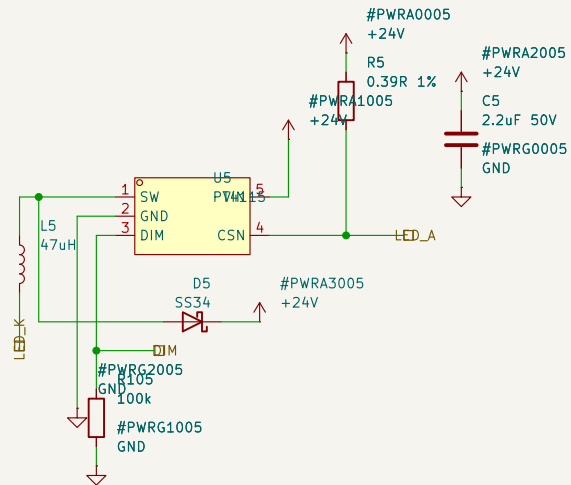
Date: 2026-07-13

Rev:

KiCad E.D.A. 10.0.3

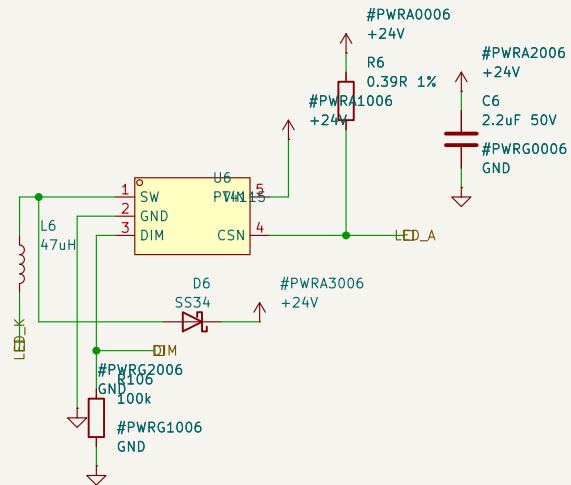
Id: 5/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH05/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 6/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH06/

File: channel.kicad_sch

Title: Channel: PT4115 buck 256mA

Size: A4

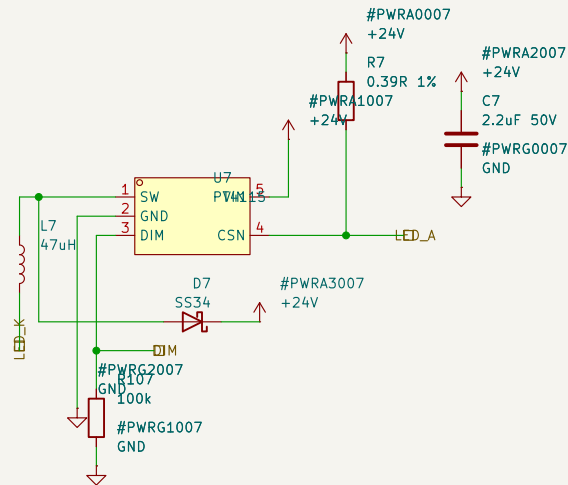
Date: 2026-07-13

Rev:

KiCad E.D.A. 10.0.3

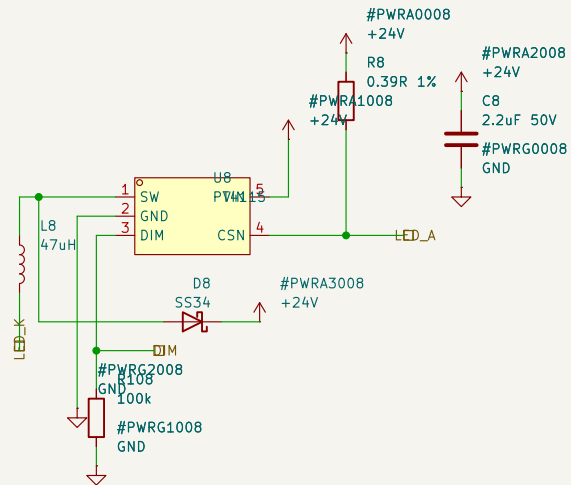
Id: 7/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



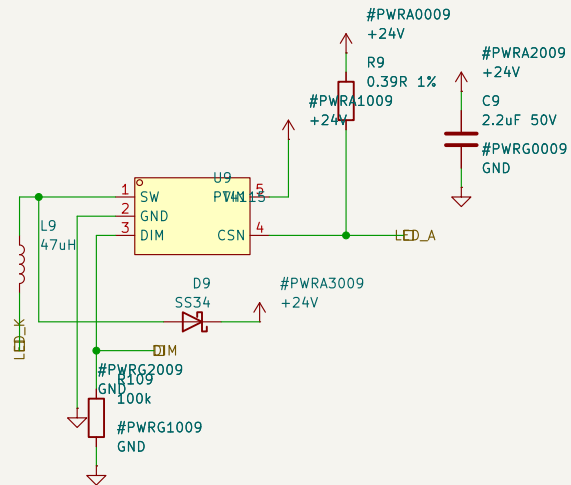
Sheet: /CH07/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3	Id: 8/97	

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH08/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 9/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH09/

File: channel.kicad_sch

Title: Channel: PT4115 buck 256mA

Size: A4

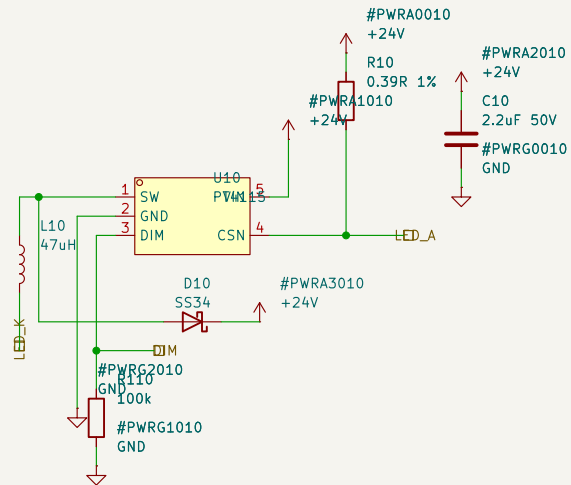
Date: 2026-07-13

Rev:

KiCad E.D.A. 10.0.3

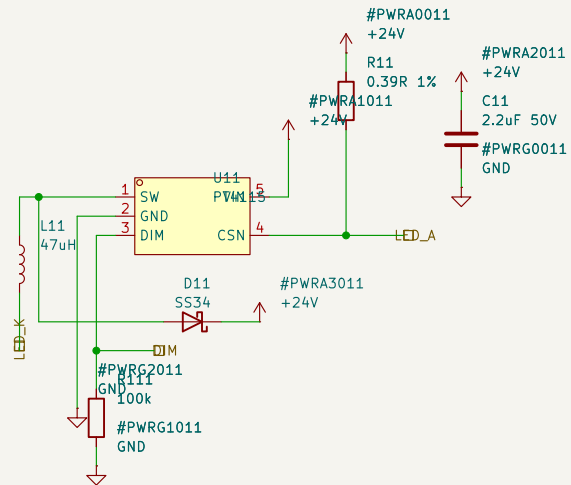
Id: 10/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



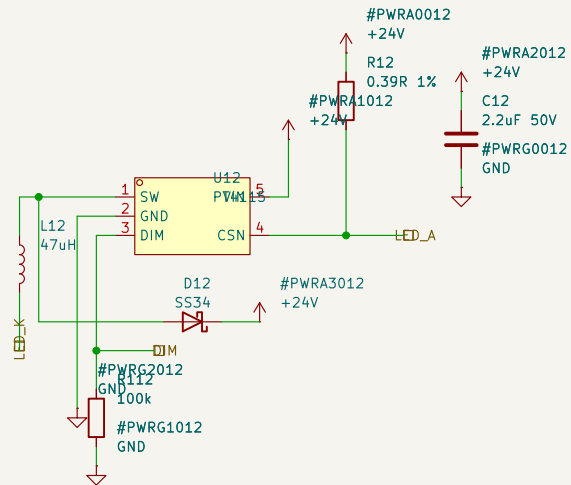
Sheet: /CH10/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 11/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



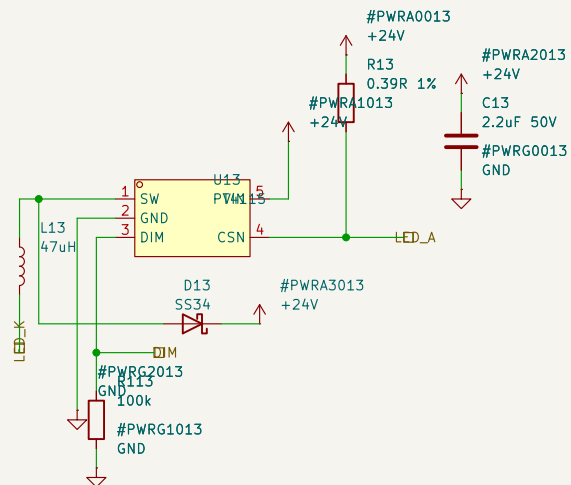
Sheet: /CH11/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 12/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



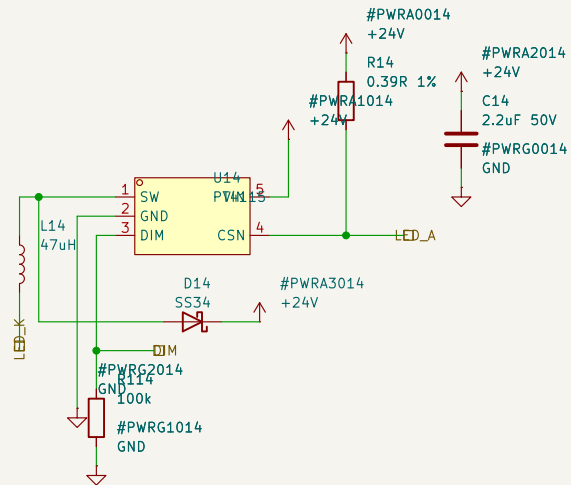
Sheet: /CH12/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 13/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



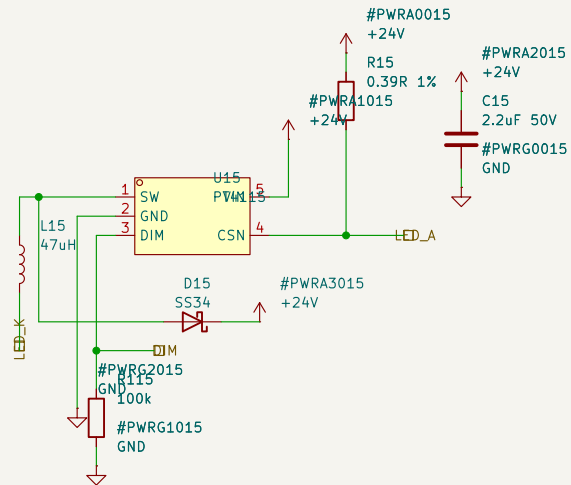
Sheet: /CH13/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 14/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH14/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 15/97

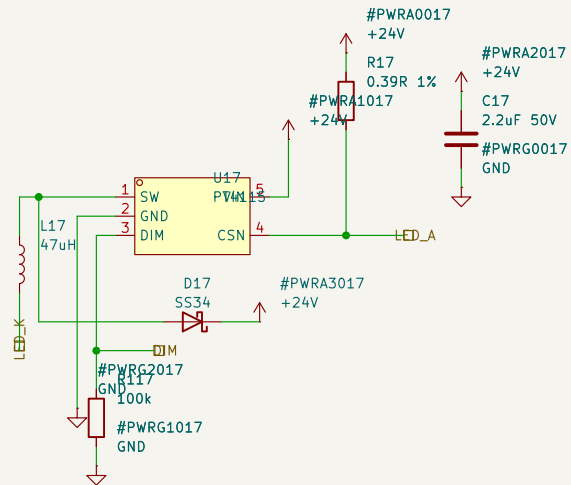
One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH15/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 16/97

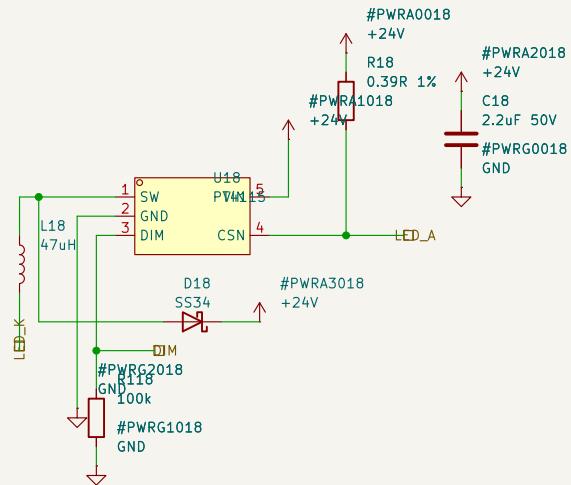
[illegible]

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH17/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3	Id: 18/97	

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH18/

File: channel.kicad_sch

Title: Channel: PT4115 buck 256mA

Size: A4

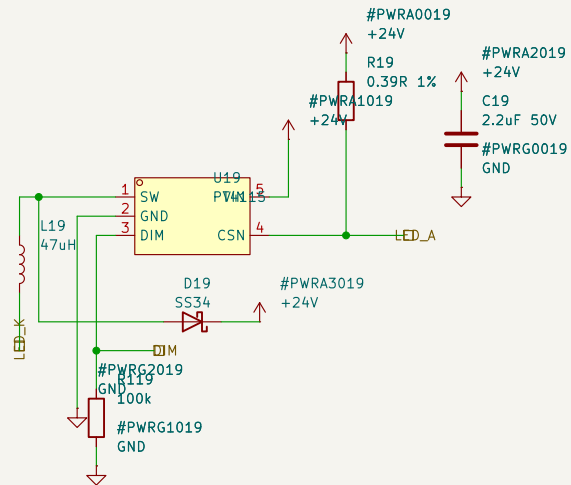
Date: 2026-07-13

Rev:

KiCad E.D.A. 10.0.3

Id: 19/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH19/

File: channel.kicad_sch

Title: Channel: PT4115 buck 256mA

Size: A4

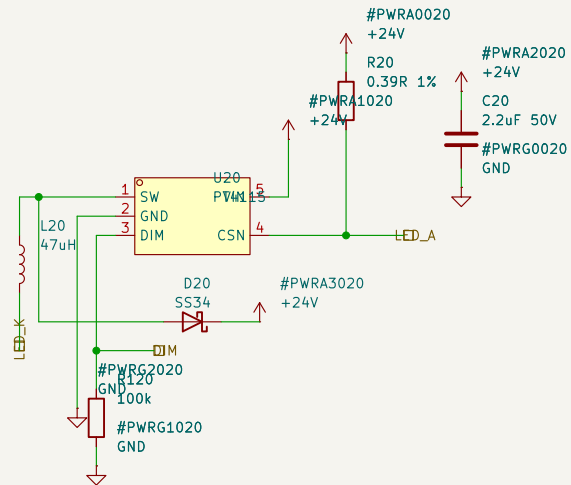
Date: 2026-07-13

Rev:

KiCad E.D.A. 10.0.3

Id: 20/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH20/

File: channel.kicad_sch

Title: Channel: PT4115 buck 256mA

Size: A4

Date: 2026-07-13

Rev:

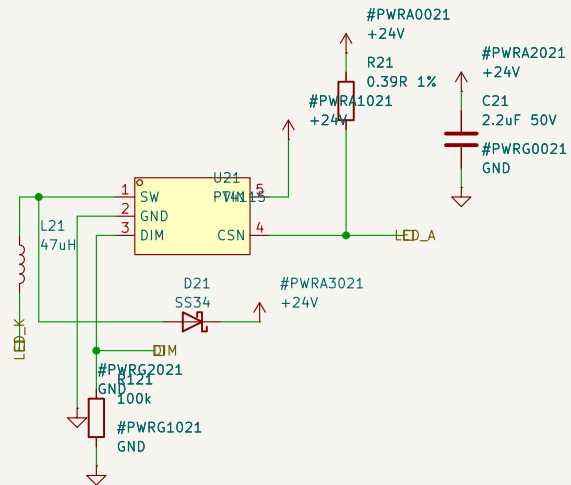
KiCad E.D.A. 10.0.3

Id: 21/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.

The schematic shows a PT4115 buck converter (U21) with pins SW, GND, DIM, CSN, PVIN, and PVOUT. The input is +24V (#PWA0021) through resistor R21 (0.39R, 1%, #PWA1021). A 47uH inductor (L21) is in series with the input. The output is LED_A, which is also connected to a 2.2uF capacitor (C21, #PWA2021) and a 50V electrolytic capacitor (#PWG0021). A diode D21 (SS34, #PWA3021) is connected from the output to ground. The DIM pin has a 100k pull-down resistor (#PWRC2021) to ground. The CSN pin has a 100k pull-up resistor (#PWRC1021) to ground. The PVIN pin is connected to ground. The PVOUT pin is connected to the output. The SW pin is connected to the input. The GND pin is connected to ground.

Sheet: /CH21/ File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 22/97



File: channel.kicad_sch

Size: A4

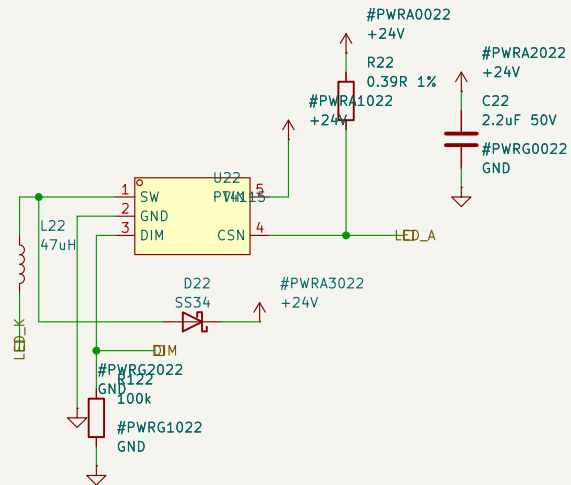
Date: 2026-07-13

SIZE: A1	5
KiCad E.D.A. 10.0.3	

Rev:

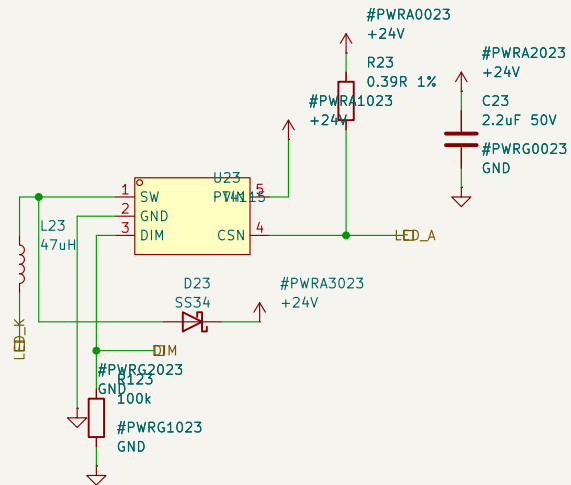
Id: 22/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



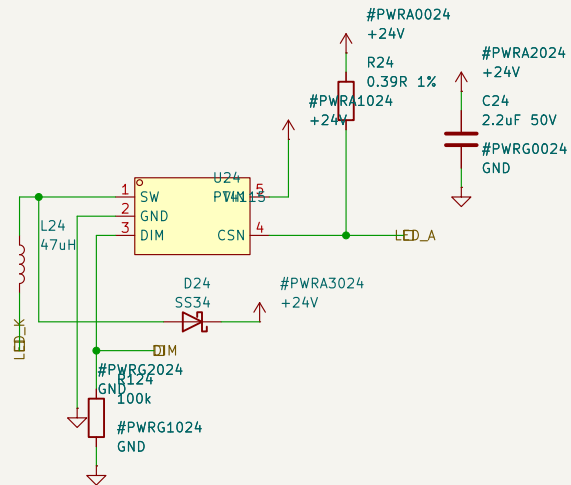
Sheet: /CH22/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 23/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



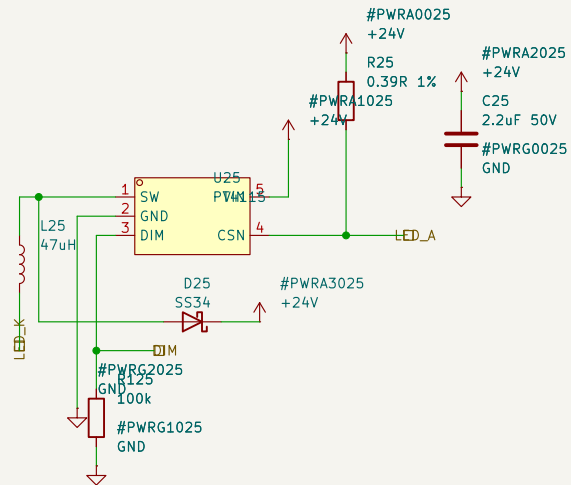
Sheet: /CH23/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 24/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH24/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 25/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH25/

File: channel.kicad_sch

Title: Channel: PT4115 buck 256mA

Size: A4

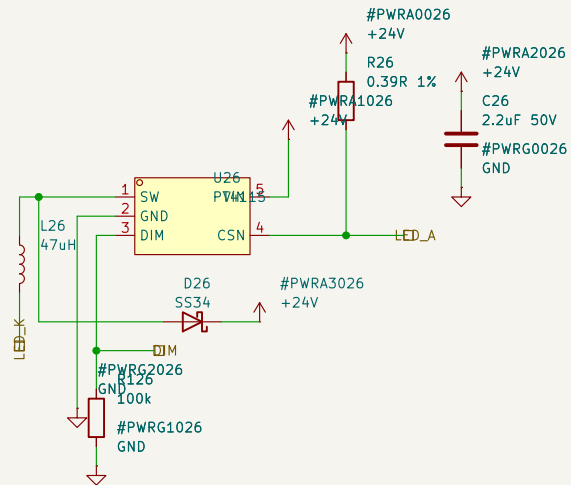
Date: 2026-07-13

Rev:

KiCad E.D.A. 10.0.3

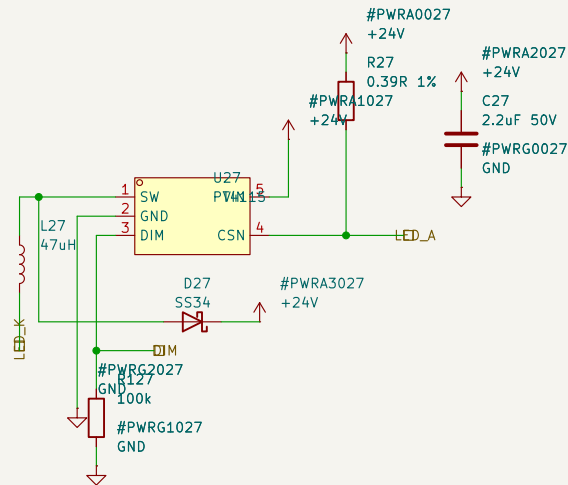
Id: 26/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



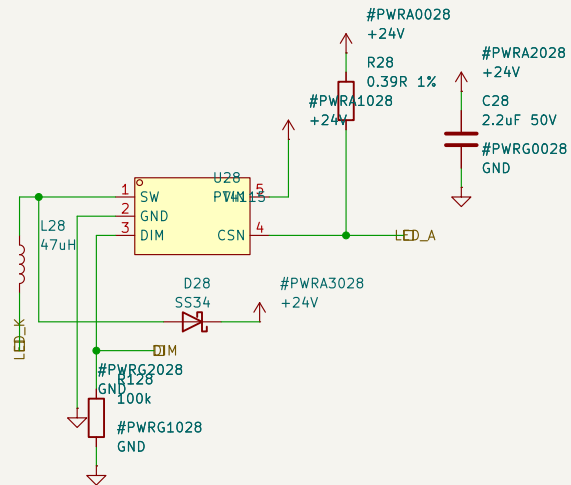
Sheet: /CH26/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 27/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



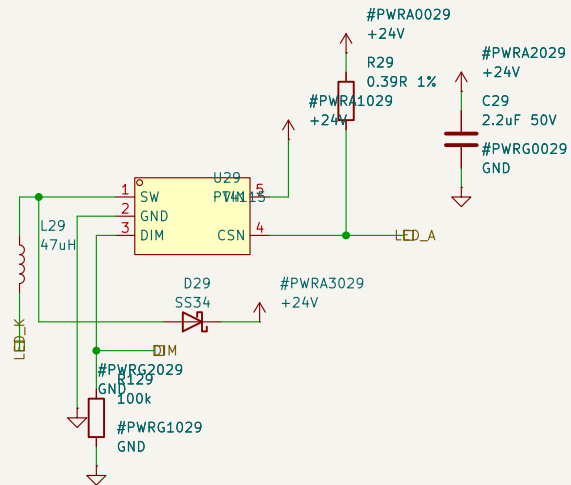
Sheet: /CH27/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 28/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



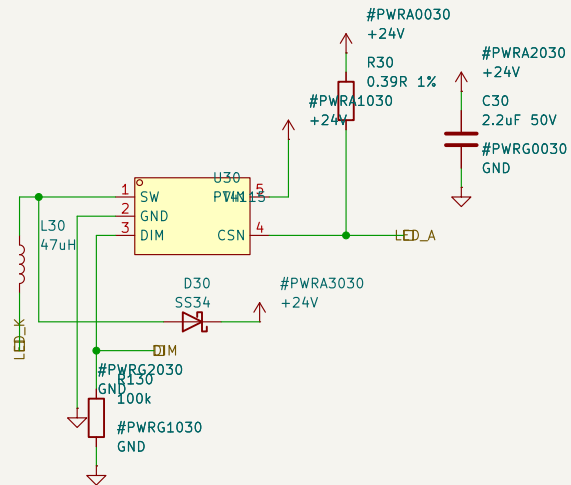
Sheet: /CH28/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 29/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH29/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 30/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH30/

File: channel.kicad_sch

Title: Channel: PT4115 buck 256mA

Size: A4

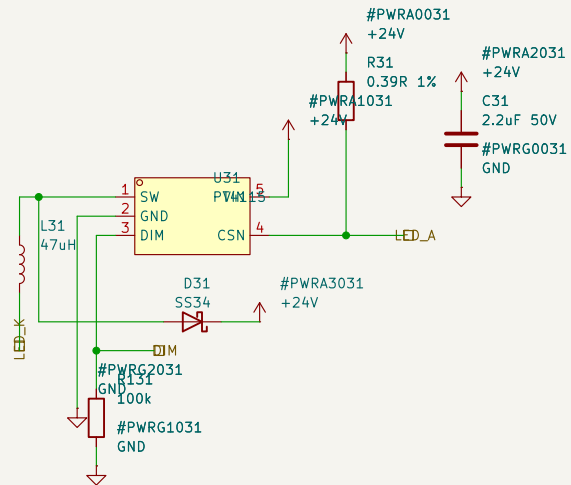
Date: 2026-07-13

Rev:

KiCad E.D.A. 10.0.3

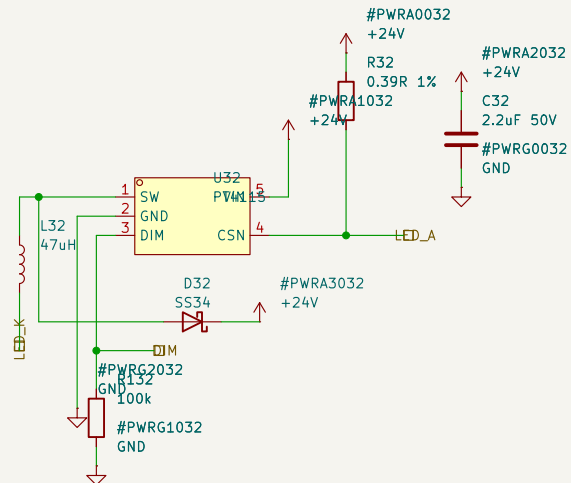
Id: 31/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



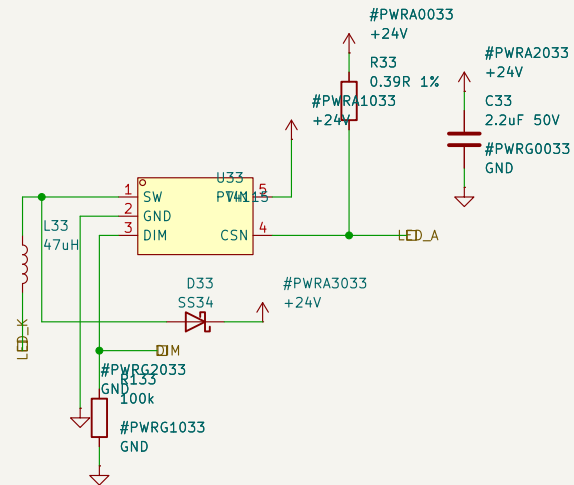
Sheet: /CH31/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 32/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



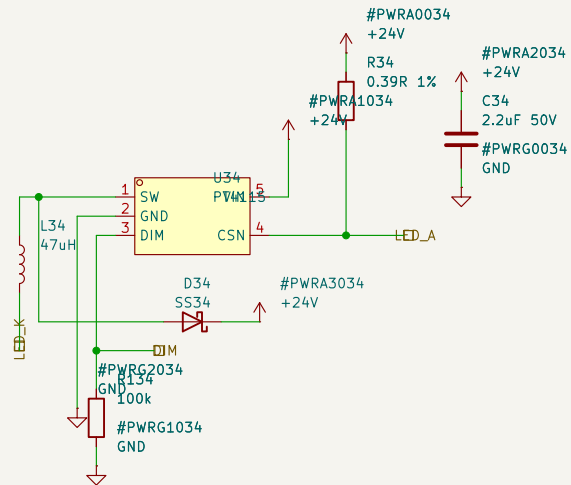
Sheet: /CH32/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 33/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH33/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 34/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH34/

File: channel.kicad_sch

Title: Channel: PT4115 buck 256mA

Size: A4

Date: 2026-07-13

Rev:

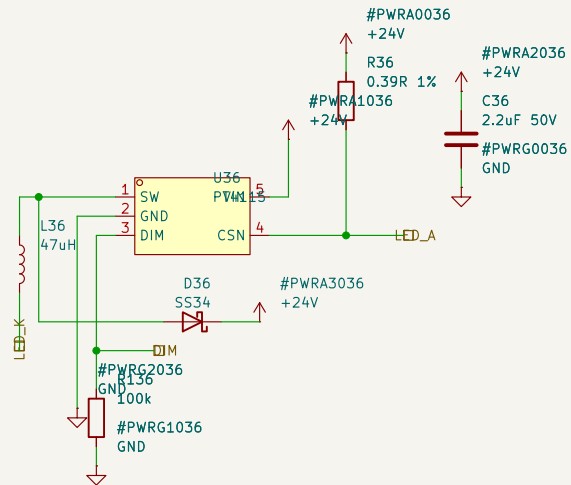
KiCad E.D.A. 10.0.3

Id: 35/97

[illegible]

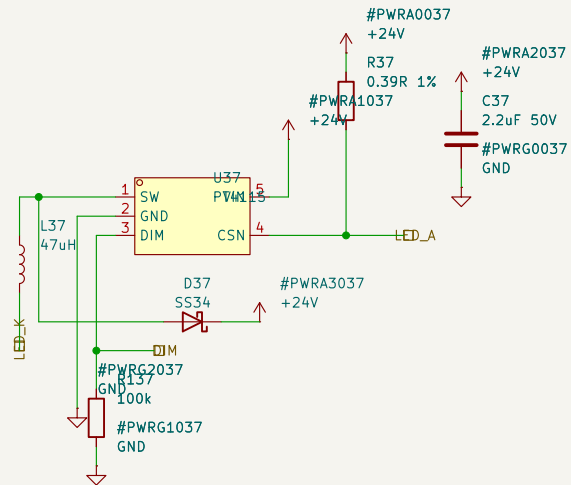
Rev:
Id: 36/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



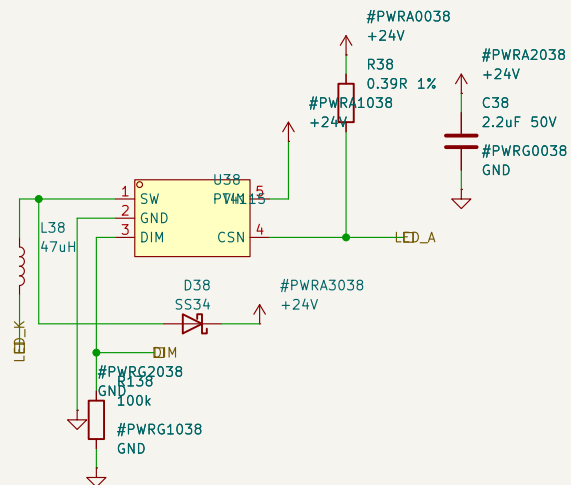
Sheet: /CH36/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 37/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



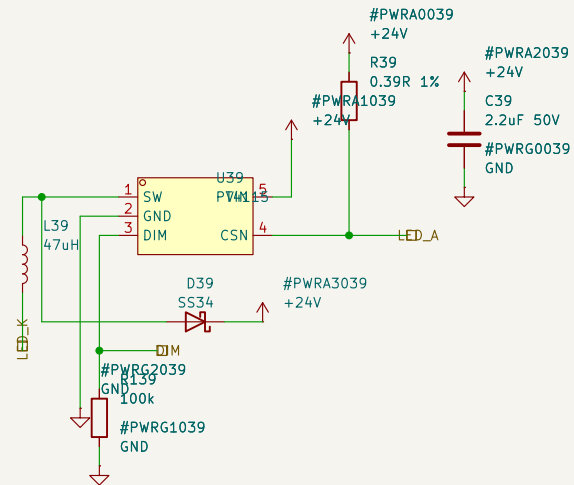
Sheet: /CH37/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 38/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH38/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 39/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH39/

File: channel.kicad_sch

Title: Channel: PT4115 buck 256mA

Size: A4

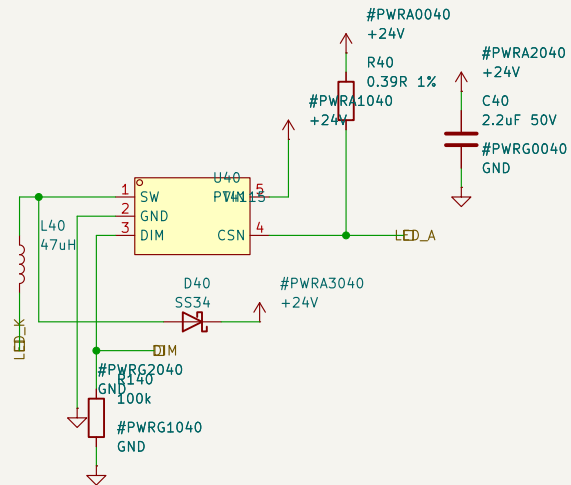
Date: 2026-07-13

Rev:

KiCad E.D.A. 10.0.3

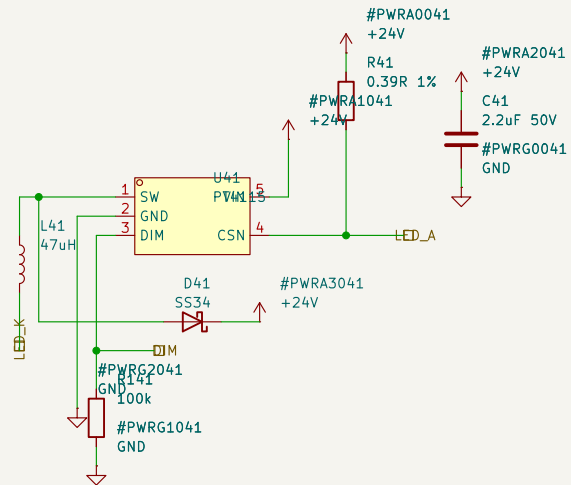
Id: 40/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



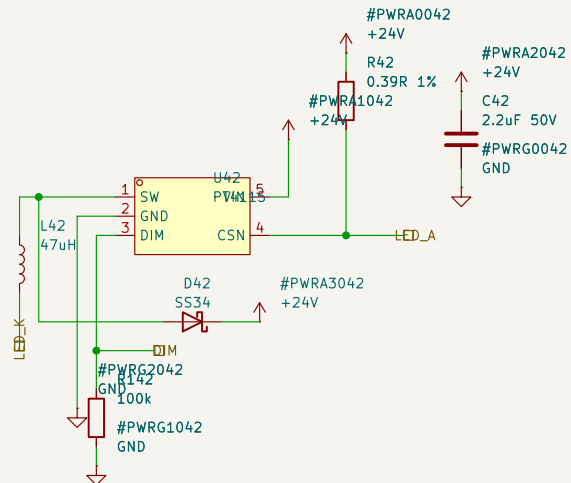
Sheet: /CH40/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 41/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH41/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 42/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH42/

File: channel.kicad_sch

Title: Channel: PT4115 buck 256mA

Size: A4

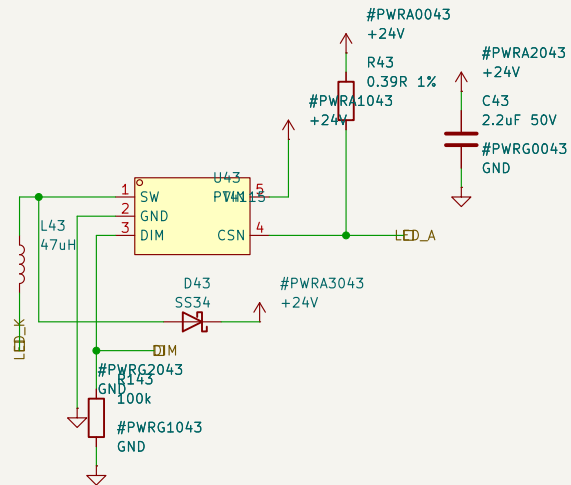
Date: 2026-07-13

Rev:

KiCad E.D.A. 10.0.3

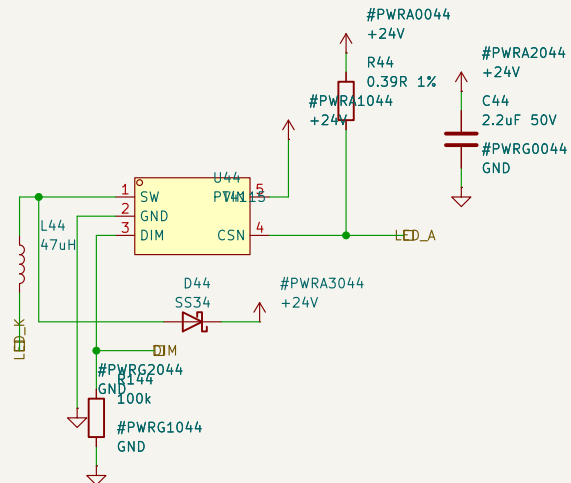
Id: 43/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH43/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 44/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH44/

File: channel.kicad_sch

Title: Channel: PT4115 buck 256mA

Size: A4

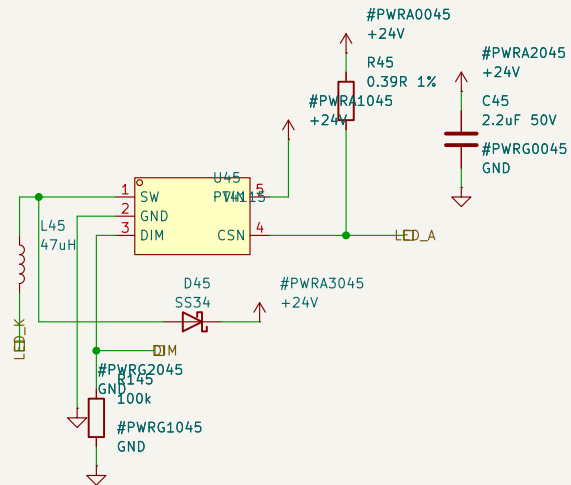
Date: 2026-07-13

Rev:

KiCad E.D.A. 10.0.3

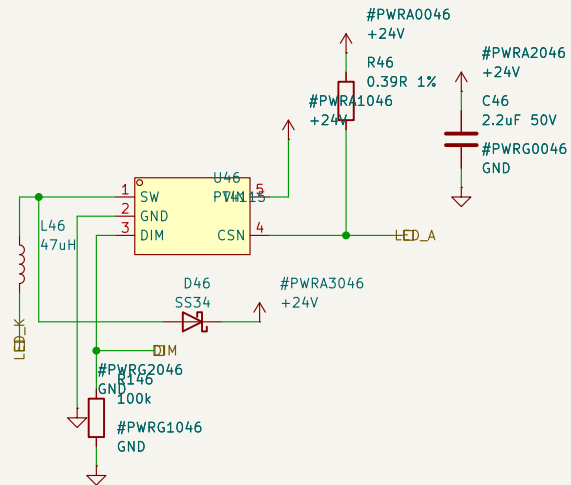
Id: 45/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH45/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 46/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH46/

File: channel.kicad_sch

Title: Channel: PT4115 buck 256mA

Size: A4

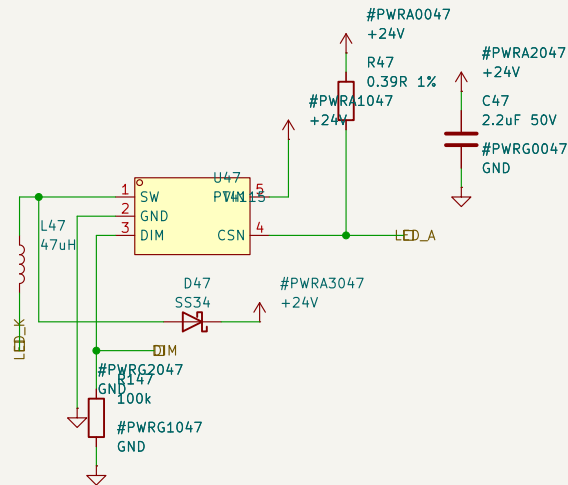
Date: 2026-07-13

Rev:

KiCad E.D.A. 10.0.3

Id: 47/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH47/

File: channel.kicad_sch

Title: Channel: PT4115 buck 256mA

Size: A4

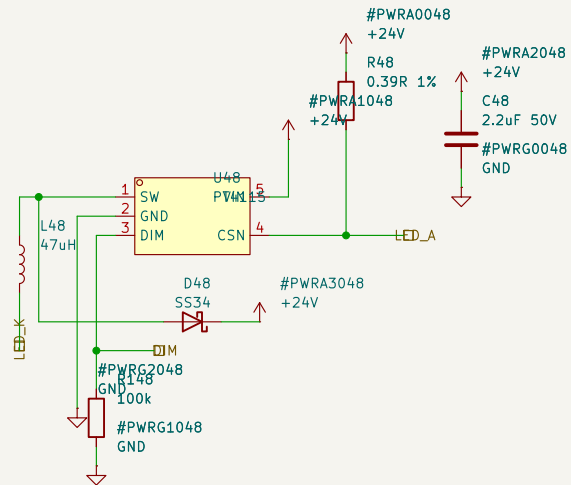
Date: 2026-07-13

Rev:

KiCad E.D.A. 10.0.3

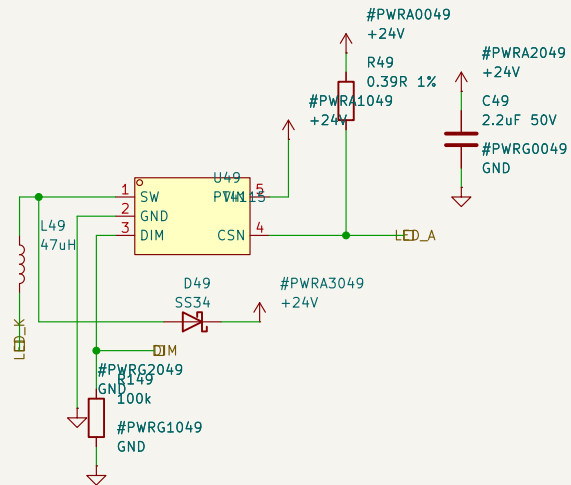
Id: 48/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH48/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 49/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH49/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 50/97

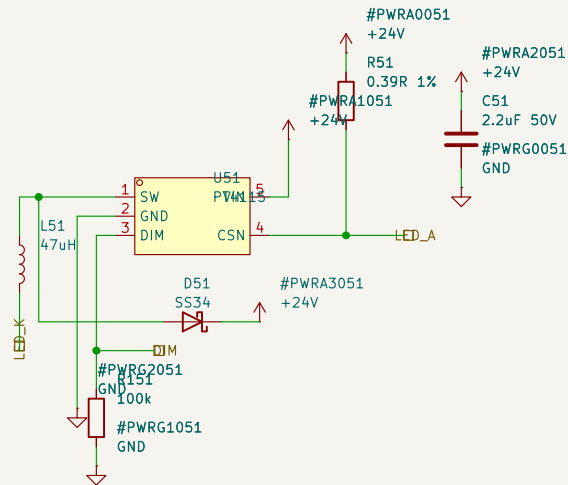
One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.

The schematic diagram shows a PT4115 buck converter (U50) circuit. The input is +24V, connected to the SW pin (pin 1) through a 47uH inductor (L50). The GND pin (pin 2) is connected to ground. The DIM pin (pin 3) is connected to ground through a 100k resistor. The CSN pin (pin 4) is connected to the LED_A output. The output is connected to the LED_A output through a 0.39R resistor (R50). A 2.2uF capacitor (C50) is connected between the output and ground. An SS34 diode (D50) is connected between the output and ground. The DIM pin is also connected to ground through a 100k resistor. The output is connected to the LED_A output through a 0.39R resistor (R50). A 2.2uF capacitor (C50) is connected between the output and ground. An SS34 diode (D50) is connected between the output and ground. The DIM pin is also connected to ground through a 100k resistor.

Sheet: /CH50/		Date: 2026-07-13		Rev:
File: channel.kicad_sch		Title: Channel: PT4115 buck 256mA		
Size: A4	KiCad E.D.A. 10.0.3			Id: 51/97

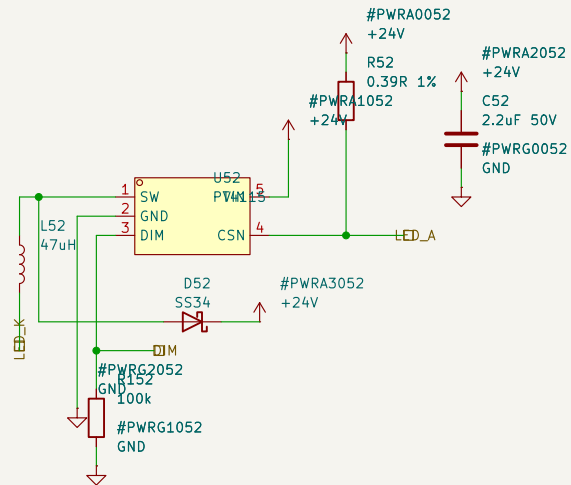
Rev:
Id: 51/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



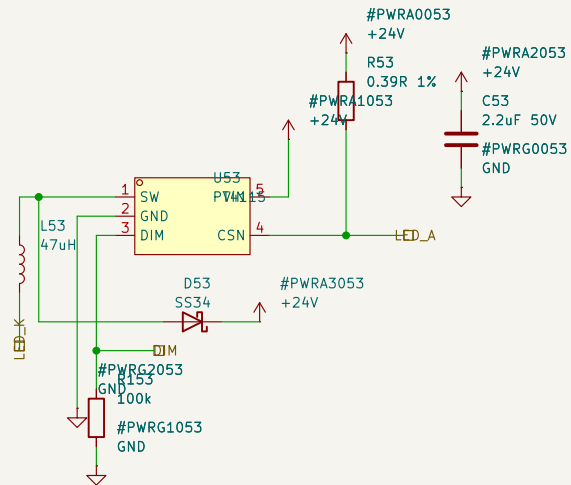
Sheet: /CH51/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 52/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



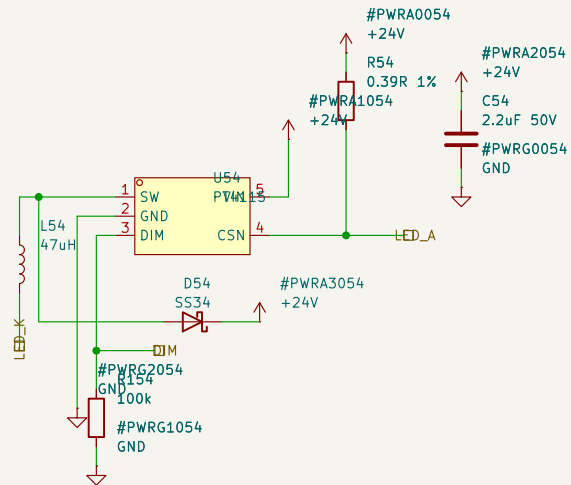
Sheet: /CH52/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 53/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH53/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3	Id: 54/97	

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



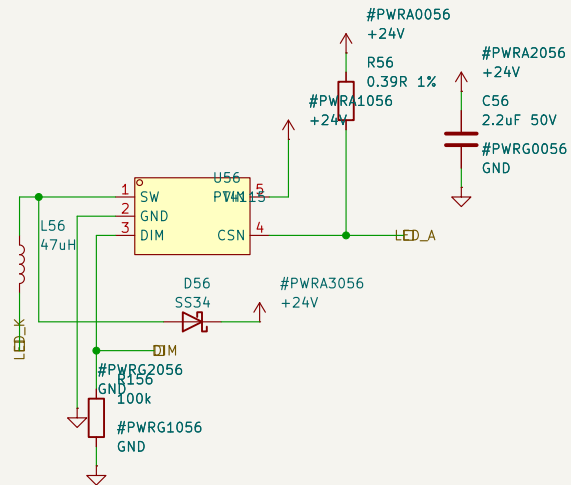
Sheet: /CH54/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 55/97

[illegible]

Rev:
Id: 56/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.

Sheet: /CH56/
File: channel.kicad_sch
Title: Channel: PT4115 buck 256mA
Size: A4 Date: 2026-07-13 Rev:
KiCad E.D.A. 10.0.3 Id: 57/97



File: channel.kicad_sch

Size: A4

Date: 2026-07-13

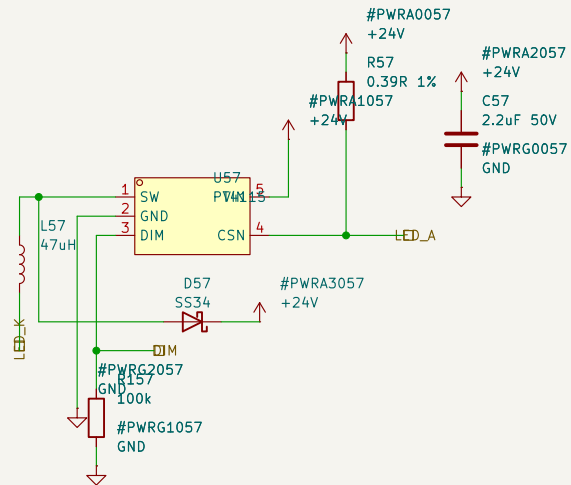
SIZE: A1	5
KiCad E.D.A. 10.0.3	

Rev:

Id: 57/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.

Sheet: /CH57/
File: channel.kicad_sch
Title: Channel: PT4115 buck 256mA
Size: A4 Date: 2026-07-13 Rev:
KiCad E.D.A. 10.0.3 Id: 58/97



File: channel.kicad_sch

Size: A4

Date: 2026-07-13

SIZE: A1	5
KiCad E.D.A. 10.0.3	

Rev:

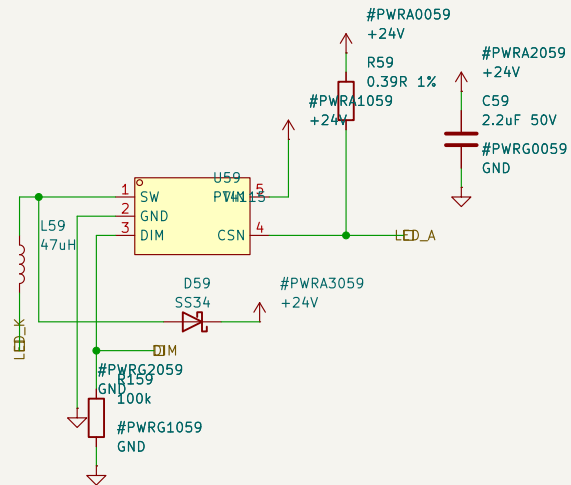
Id: 58/97

A
B
C
D



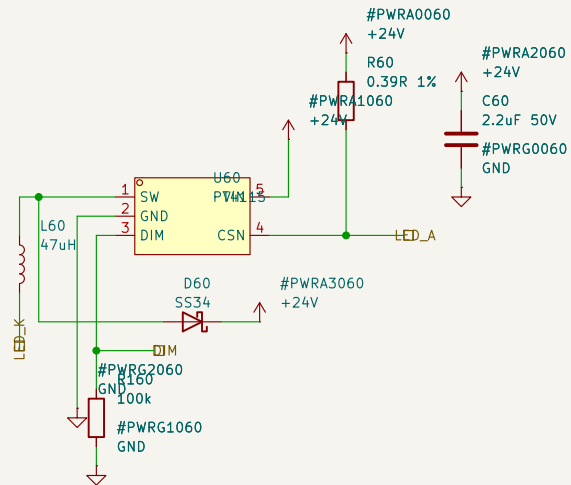
Id: 59/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



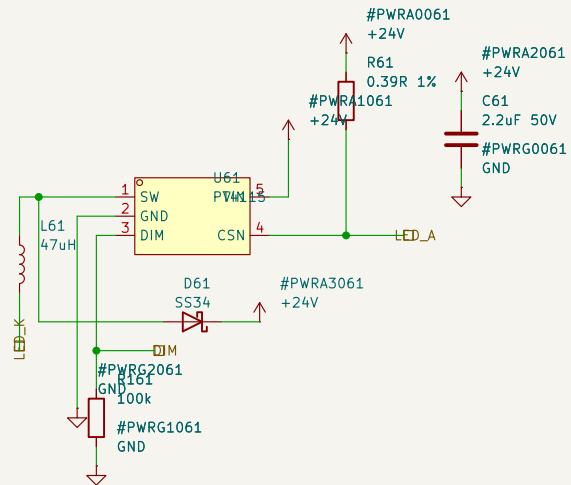
Sheet: /CH59/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 60/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH60/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 61/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.

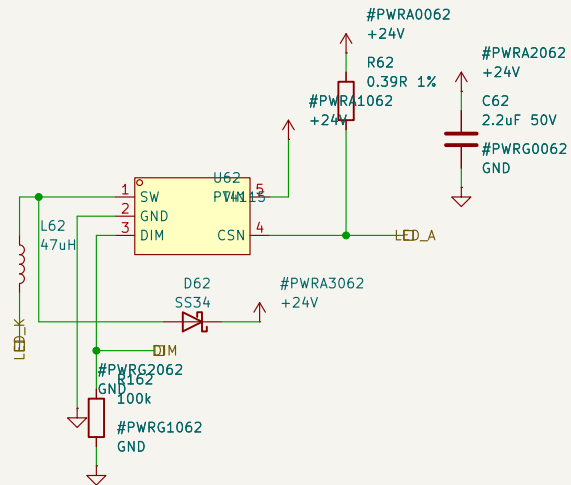


Sheet: /CH61/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 62/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.

The schematic shows a PT4115 buck converter (U62) used as a constant current source. The input is +24V (#PWA0062). A sense resistor R62 (0.39R, 1%) is placed between the input and the SW pin (pin 1). An inductor L62 (47uH) is connected between the SW pin and the LED_K pin. The GND pin (pin 2) is connected to ground (#PWGR2062). The DIM pin (pin 3) has a 100k pull-down resistor (#PWGR1062) to ground. The CSN pin (pin 4) is connected to the LED_A pin. The output of the LED is connected to the ribbon cable. A diode D62 (SS34) is connected from the LED_K pin to the +24V rail. A capacitor C62 (2.2uF, 50V) is connected from the +24V rail to ground (#PWGR0062).

Sheet: /CH62/ File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 63/97



File: channel.kicad_sch

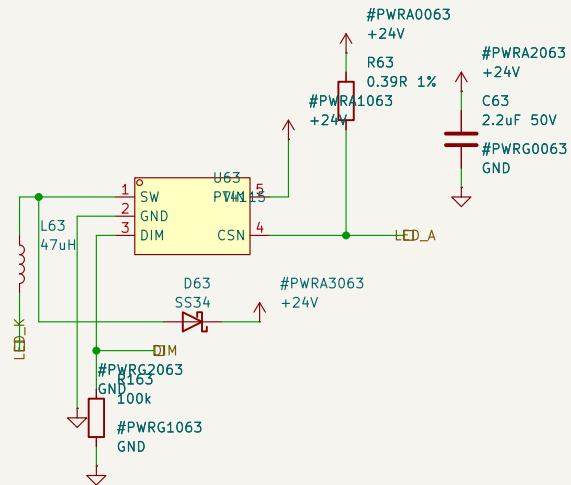
Size: A4

Date: 2026-07-13

Rev:

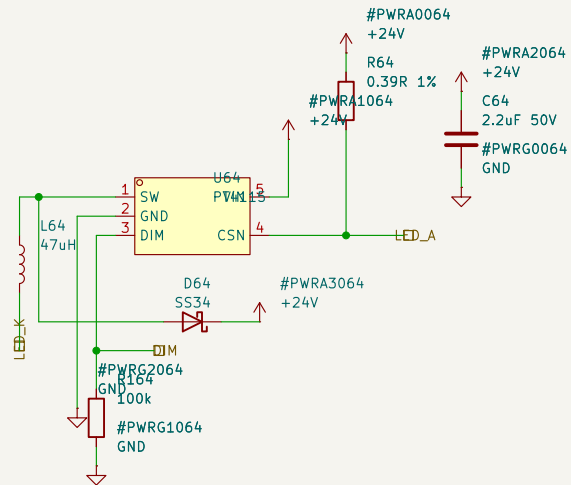
Id: 63/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



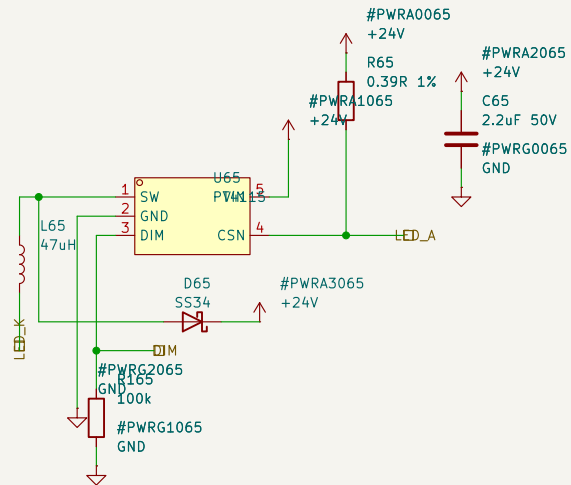
Sheet: /CH63/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 64/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



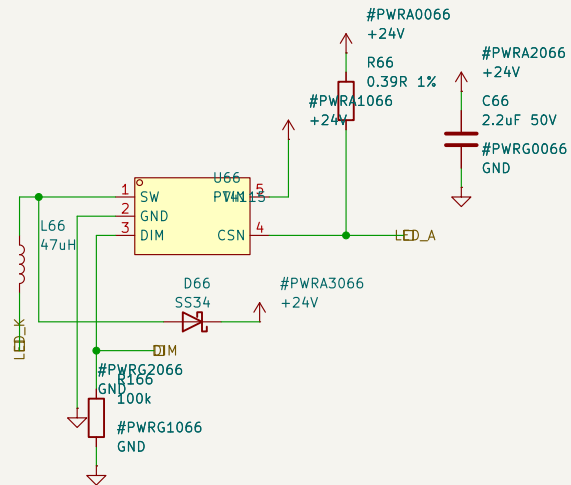
Sheet: /CH64/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 65/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



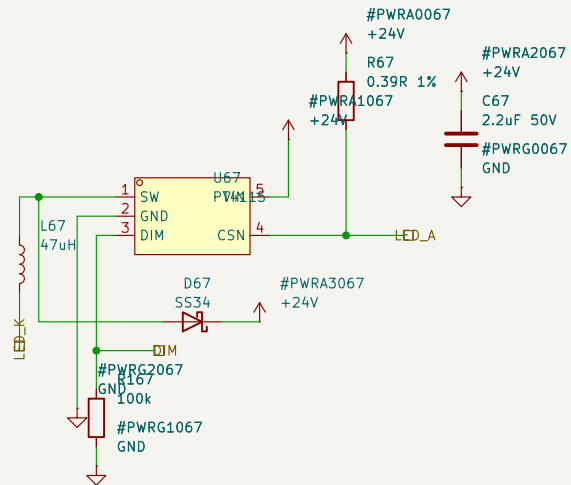
Sheet: /CH65/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 66/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



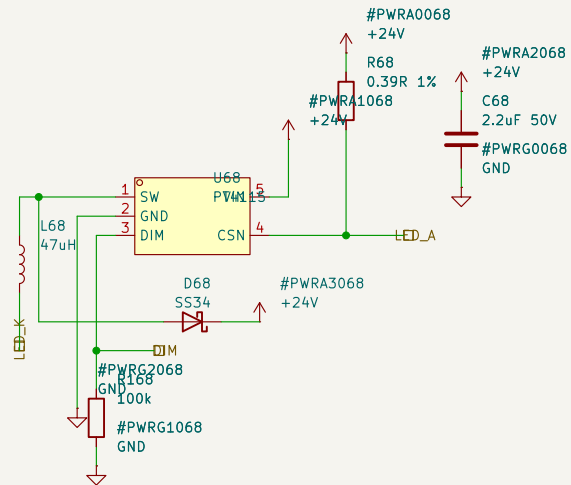
Sheet: /CH66/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 67/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH67/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 68/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH68/

File: channel.kicad_sch

Title: Channel: PT4115 buck 256mA

Size: A4

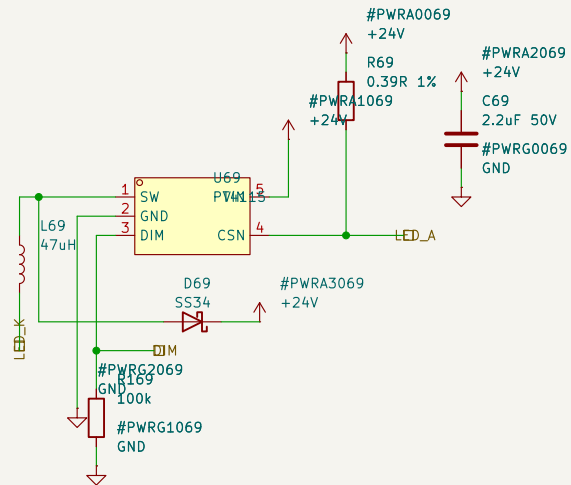
Date: 2026-07-13

Rev:

KiCad E.D.A. 10.0.3

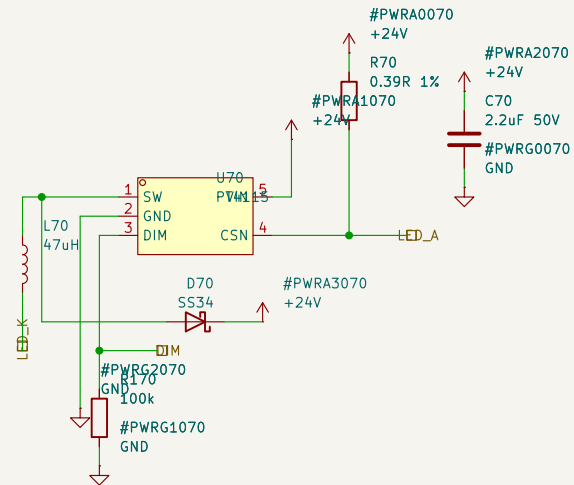
Id: 69/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



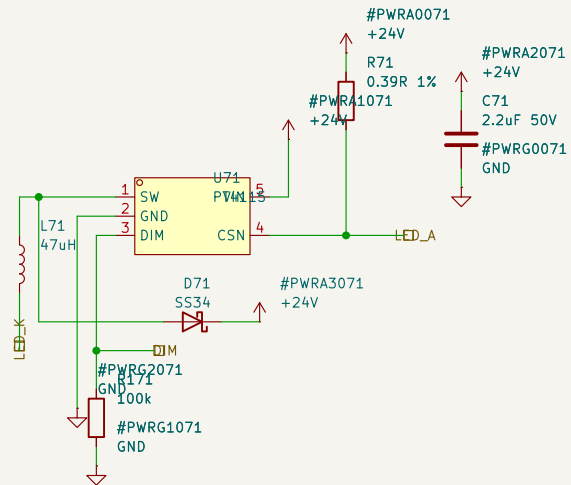
Sheet: /CH69/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 70/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH70/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 71/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH71/

File: channel.kicad_sch

Title: Channel: PT4115 buck 256mA

Size: A4

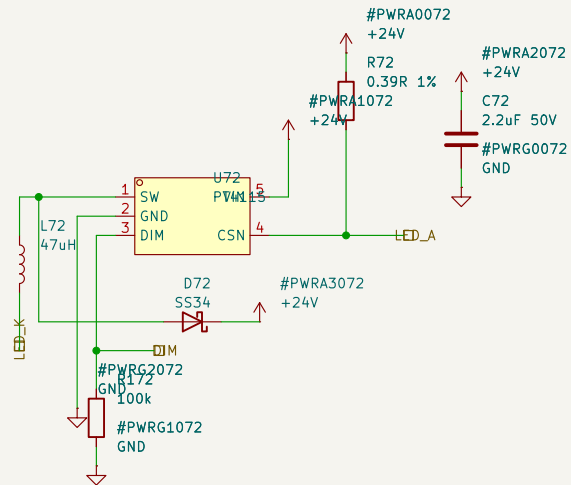
Date: 2026-07-13

Rev:

KiCad E.D.A. 10.0.3

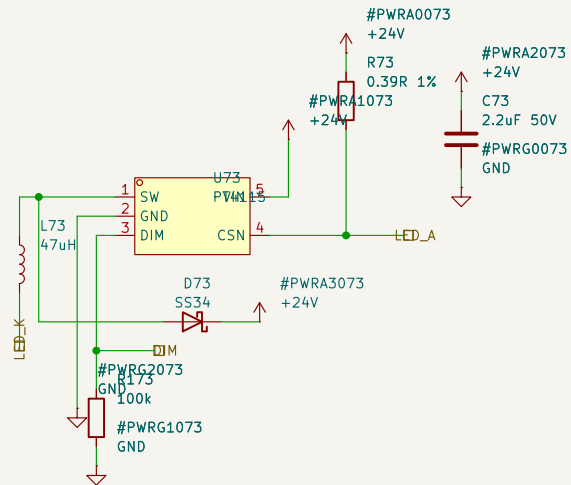
Id: 72/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



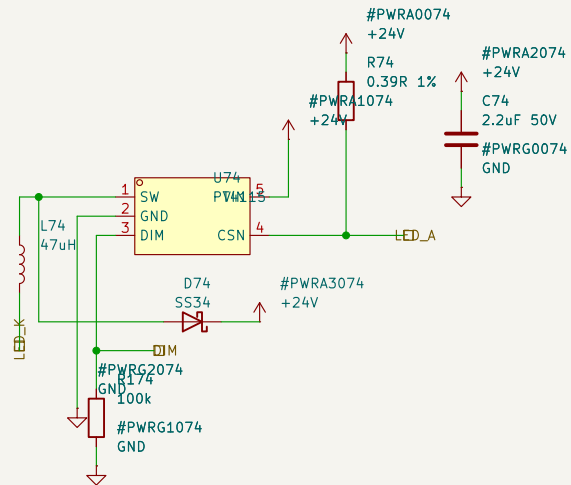
Sheet: /CH72/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 73/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



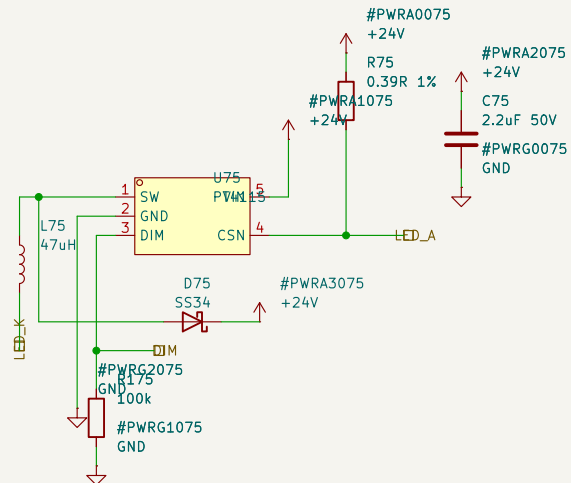
Sheet: /CH73/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 74/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH74/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 75/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH75/

File: channel.kicad_sch

Title: Channel: PT4115 buck 256mA

Size: A4

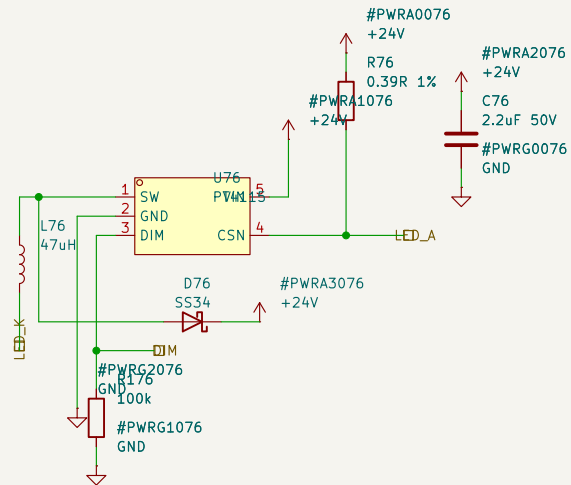
Date: 2026-07-13

Rev:

KiCad E.D.A. 10.0.3

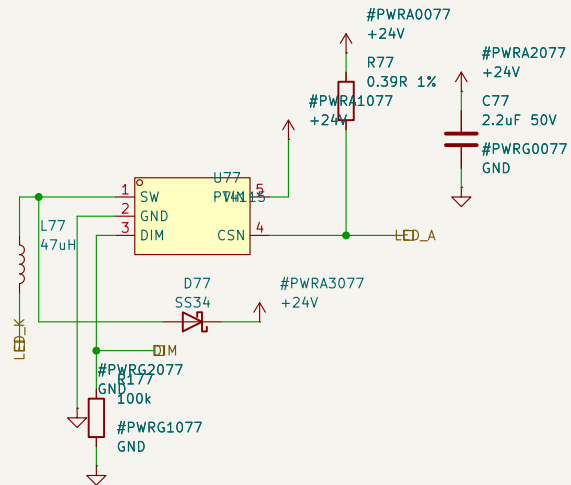
Id: 76/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



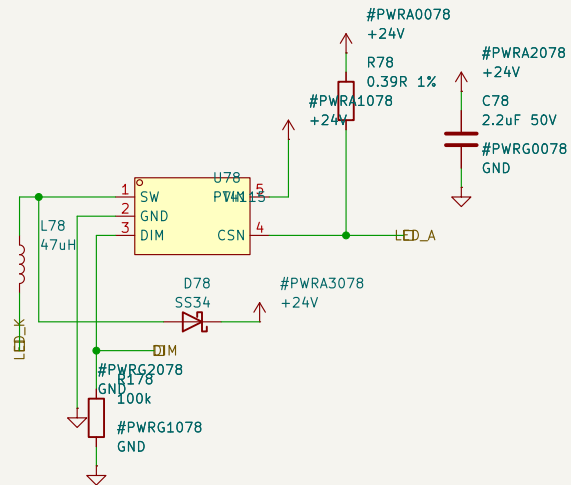
Sheet: /CH76/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 77/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH77/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3	Id: 78/97	

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH78/

File: channel.kicad_sch

Title: Channel: PT4115 buck 256mA

Size: A4

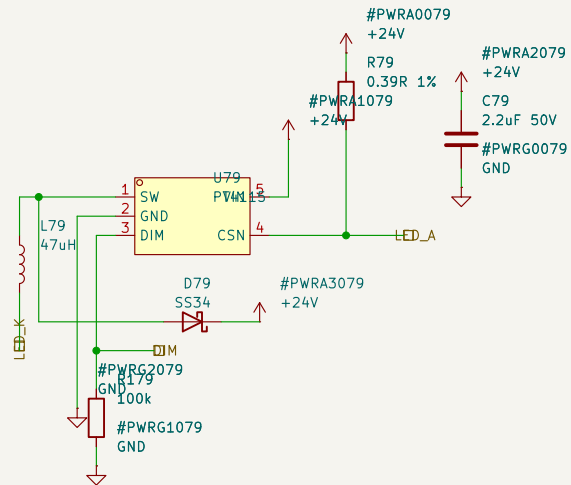
Date: 2026-07-13

Rev:

KiCad E.D.A. 10.0.3

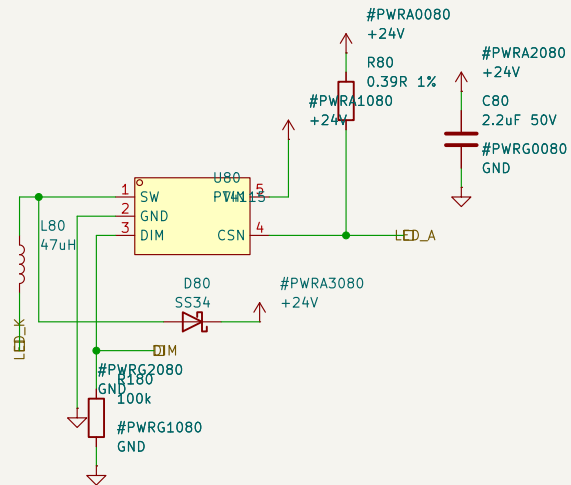
Id: 79/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



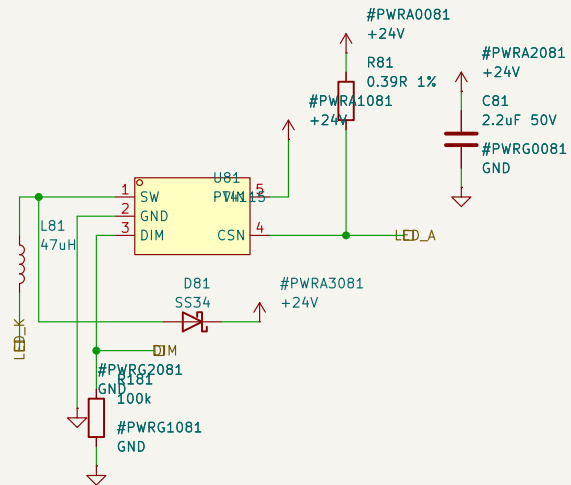
Sheet: /CH79/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 80/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



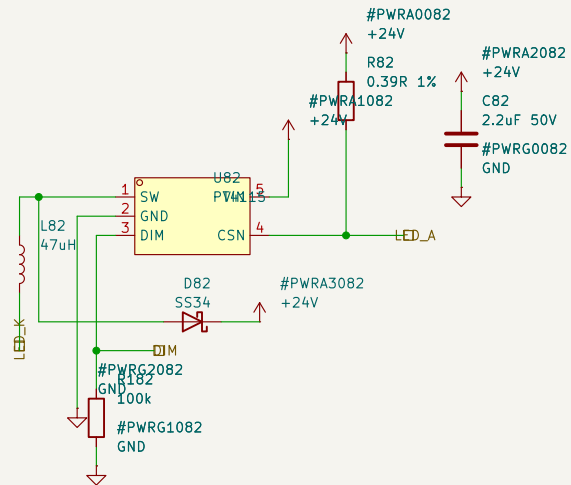
Sheet: /CH80/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3	Id: 81/97	

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



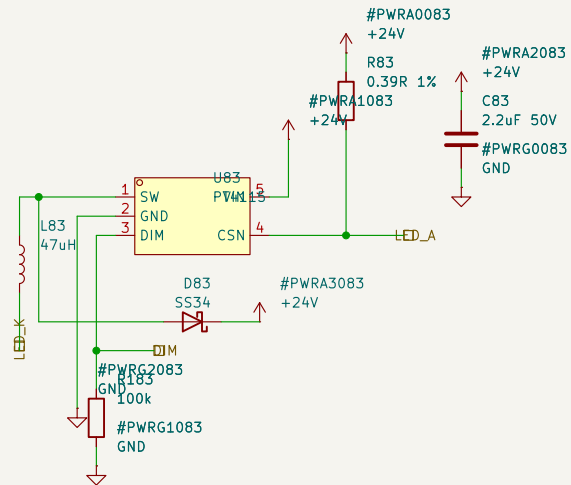
Sheet: /CH81/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 82/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH82/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 83/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH83/

File: channel.kicad_sch

Title: Channel: PT4115 buck 256mA

Size: A4

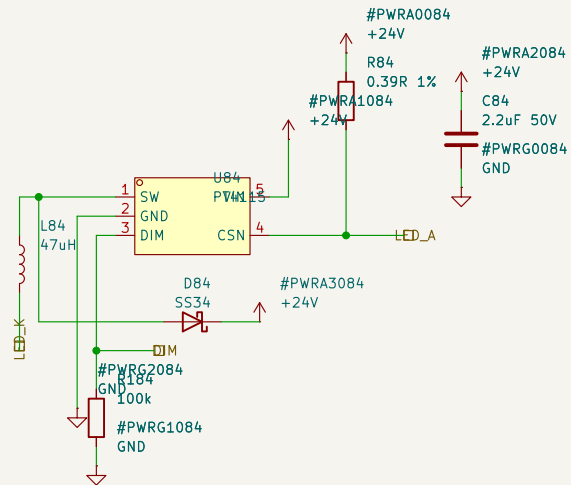
Date: 2026-07-13

Rev:

KiCad E.D.A. 10.0.3

Id: 84/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH84/

File: channel.kicad_sch

Title: Channel: PT4115 buck 256mA

Size: A4

Date: 2026-07-13

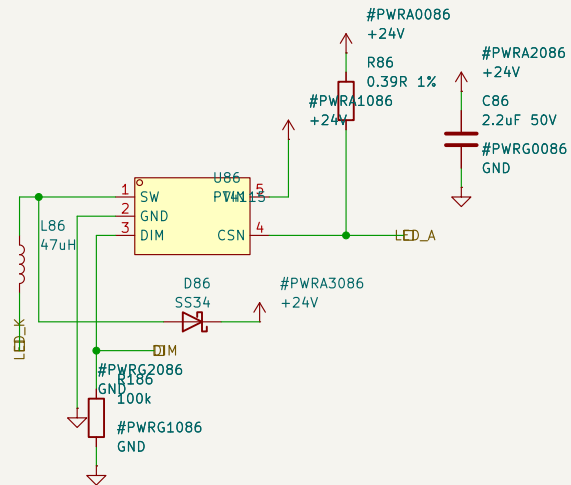
Rev:

KiCad E.D.A. 10.0.3

Id: 85/97

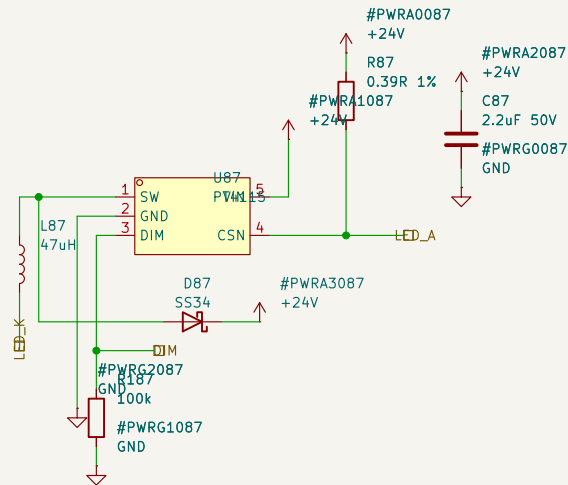
[illegible]

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



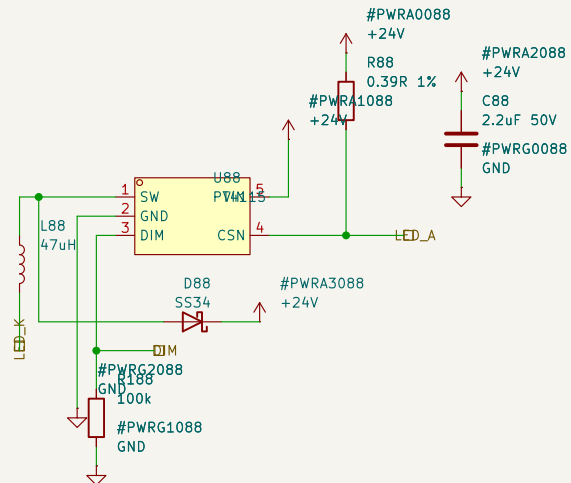
Sheet: /CH86/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 87/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH87/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 88/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH88/

File: channel.kicad_sch

Title: Channel: PT4115 buck 256mA

Size: A4

Date: 2026-07-13

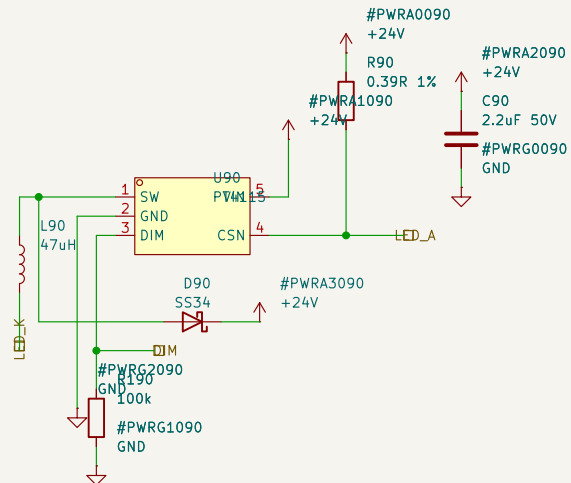
Rev:

KiCad E.D.A. 10.0.3

Id: 89/97

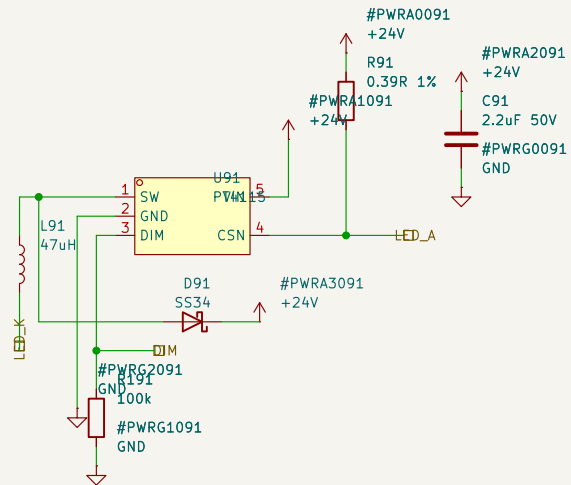
[illegible]

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



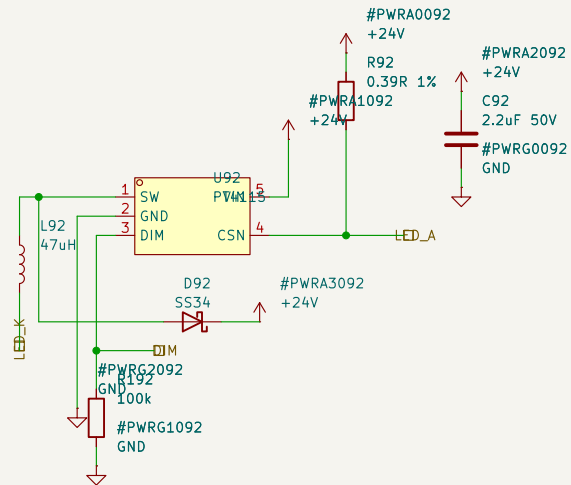
Sheet: /CH90/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 91/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH91/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 92/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH92/

File: channel.kicad_sch

Title: Channel: PT4115 buck 256mA

Size: A4

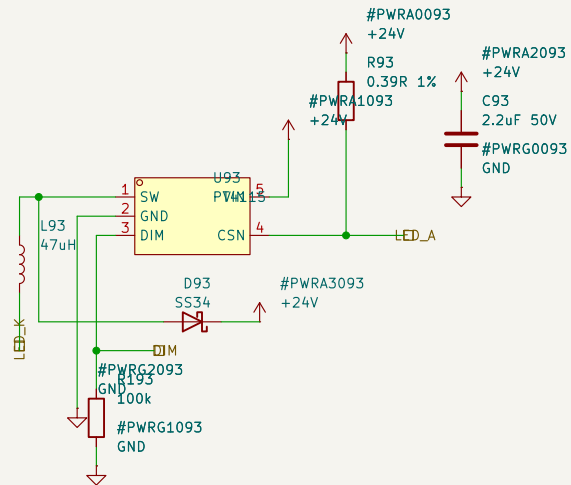
Date: 2026-07-13

Rev:

KiCad E.D.A. 10.0.3

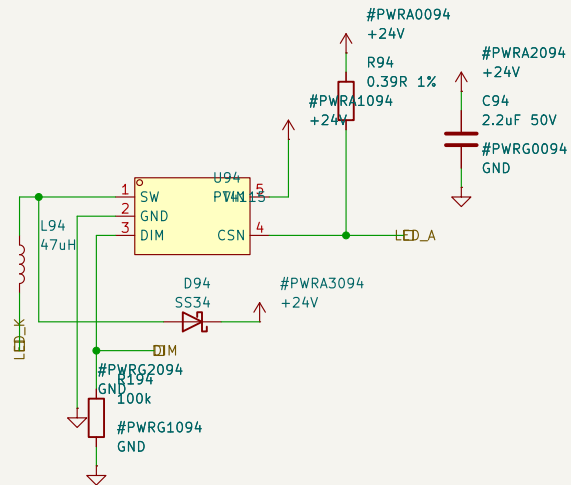
Id: 93/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



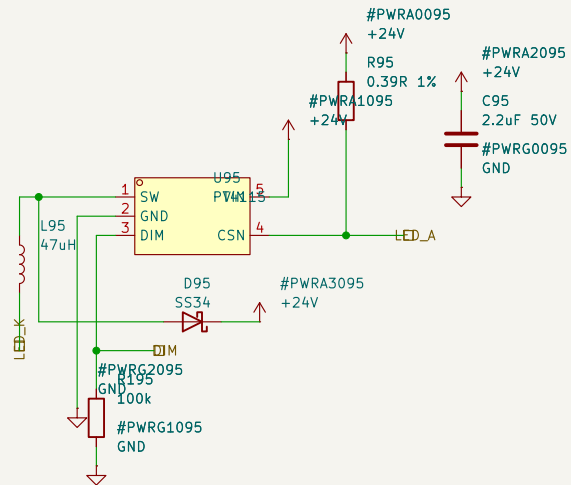
Sheet: /CH93/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 94/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



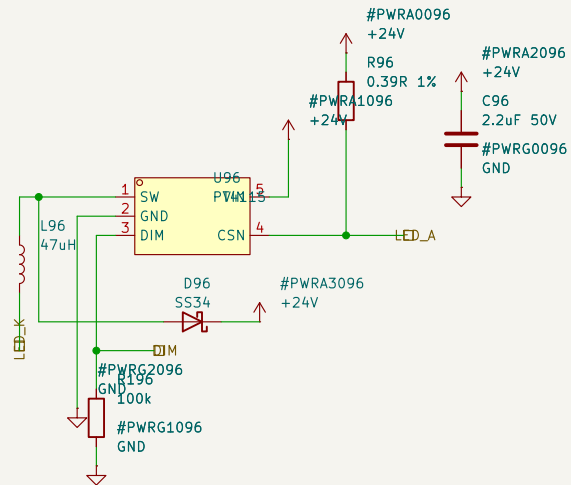
Sheet: /CH94/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 95/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH95/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 96/97

One channel: PT4115 buck CC @ 256mA ($R_s=0.39R$). SS34 band to +24V. DIM: 100k pulldown = dark boot.
LED_A/LED_K go to the ribbon; wire crossings without junction dots are NOT connected.



Sheet: /CH96/		
File: channel.kicad_sch		
Title: Channel: PT4115 buck 256mA		
Size: A4	Date: 2026-07-13	Rev:
KiCad E.D.A. 10.0.3		Id: 97/97